SPHOORTHY ENGINEERING COLLEGE | *(UGC Autonomous — NAAC Grade A)*

Structural Analysis & Design of Concrete Structures

Department: CIVIL	**Course / Subject:** Structural Analysis (CE501PC)
Semester: Semester 5	**Regulation:** Autonomous (SR23 / SR24)

Course Overview & Key Focus: Static and kinematic indeterminacy, Moment Distribution Method, Slope Deflection Method, Limit State Design of singly/doubly reinforced beams, and Slab detailing.

Unit 1: Energy Theorems & Indeterminacy

Strain energy due to axial load, bending, torsion, Castigliano's first and second theorems, Unit load method for deflection of determinate trusses.

Unit 2: Slope Deflection Method

Slope-deflection equations formulation, Continuous beams with and without support settlement, Portal frames analysis without sidesway.

Unit 3: Moment Distribution Method

Hardy Cross method, Stiffness factor, Carry-over factor, Distribution factor, Analysis of continuous beams with overhangs and fixed supports.

Unit 4: Limit State Method of Concrete Design

IS 456:2000 design philosophy, Characteristic strength, Stress block parameters, Analysis of singly reinforced and doubly reinforced rectangular RC beams.

Unit 5: Design of Beams & Slabs

Shear design, Development length, Curtailment of reinforcement, Design of one-way spanning slabs, Design of two-way slabs with torsion reinforcement at corners.